



- 7 An isolated stationary nucleus Q decays into nucleus R and an  $\alpha$ -particle. The  $\alpha$ -particle has speed  $1.5 \times 10^7 \text{ ms}^{-1}$ .

(a) Complete the equation for this decay.



[1]

(b) By considering momentum, calculate the speed of nucleus R after the decay.

speed = .....  $\text{ms}^{-1}$  [3]

(c) State **three** quantities that are conserved during the decay.

- 1 .....
- 2 .....
- 3 .....

[3]

[Total: 7]

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